

```
In [4]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
```

```
In [8]: data = {
    "Temperature": [32,30,35,29,31,33,34,28,36,27],
    "Humidity": [60,58,65,55,61,63,64,54,67,52],
    "WindSpeed": [12,10,15,9,11,13,14,8,16,7],
    "EnergyConsumption": [450,430,480,410,445,460,470,400,490,390]
}

df = pd.DataFrame(data)

df.to_csv("energy_consumption.csv", index=False)

print("Dataset created successfully!")
```

Dataset created successfully!

```
In [9]: print(df.info())

print(df.describe())

print(df.isnull().sum())
```

```
<class 'pandas.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Temperature           10 non-null    int64
1   Humidity               10 non-null    int64
2   WindSpeed              10 non-null    int64
3   EnergyConsumption      10 non-null    int64
dtypes: int64(4)
memory usage: 452.0 bytes
None
```

	Temperature	Humidity	WindSpeed	EnergyConsumption
count	10.00000	10.000000	10.00000	10.000000
mean	31.50000	59.900000	11.50000	442.500000
std	3.02765	5.043147	3.02765	34.258008
min	27.00000	52.000000	7.00000	390.000000
25%	29.25000	55.750000	9.25000	415.000000
50%	31.50000	60.500000	11.50000	447.500000
75%	33.75000	63.750000	13.75000	467.500000
max	36.00000	67.000000	16.00000	490.000000
Temperature	0			
Humidity	0			
WindSpeed	0			
EnergyConsumption	0			

```
dtype: int64
```

```
In [13]: X = df[['Temperature', 'Humidity', 'WindSpeed']]

y = df['EnergyConsumption']
```

```
In [14]: X_train, X_test, y_train, y_test = train_test_split(
        X,
        y,
        test_size=0.2,
        random_state=42
    )
```

```
In [15]: model = LinearRegression()

        model.fit(X_train, y_train)
```

```
Out[15]: ▼ LinearRegression ⓘ ?
        ► Parameters
```

```
In [16]: y_pred = model.predict(X_test)
```

```
In [17]: print("MAE :", mean_absolute_error(y_test, y_pred))

        print("MSE :", mean_squared_error(y_test, y_pred))

        print("R2 Score :", r2_score(y_test, y_pred))
```

```
MAE : 2.9278219971056103
MSE : 8.612098243175954
R2 Score : 0.9904310019520267
```

```
In [18]: plt.figure(figsize=(8,5))

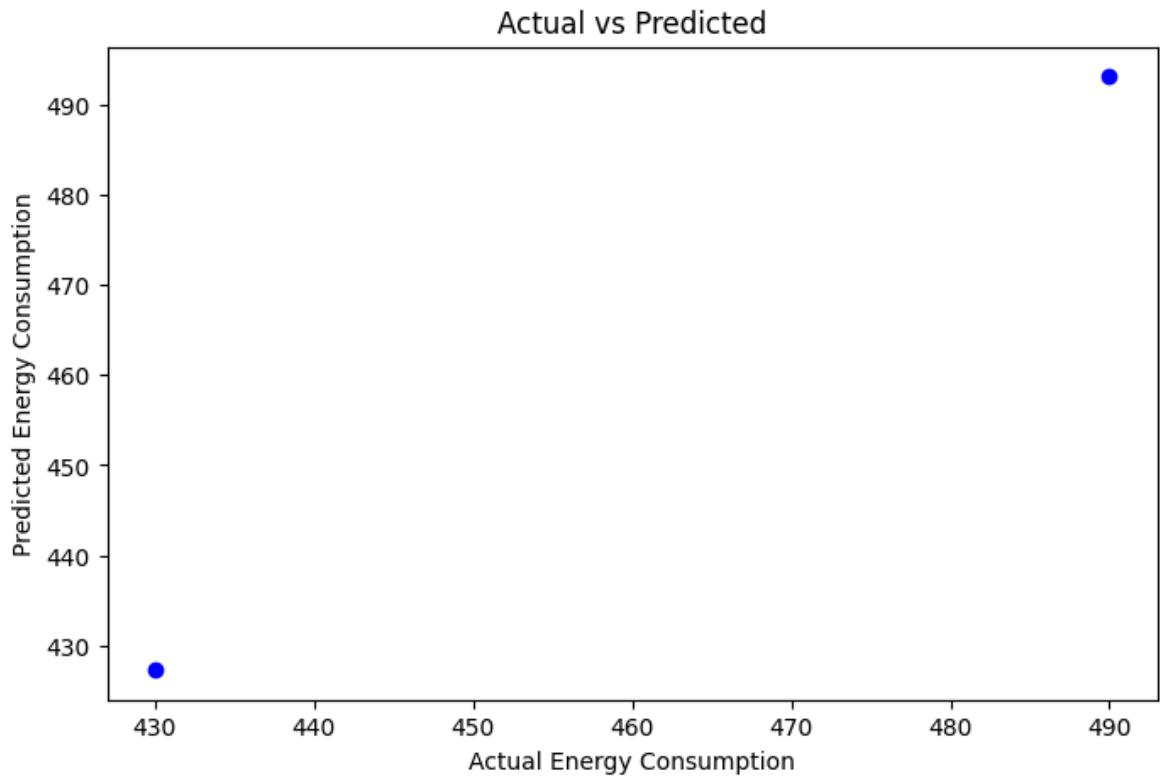
        plt.scatter(y_test, y_pred, color='blue')

        plt.xlabel("Actual Energy Consumption")

        plt.ylabel("Predicted Energy Consumption")

        plt.title("Actual vs Predicted")

        plt.show()
```



```
In [19]: new_data = pd.DataFrame({
          'Temperature':[33],
          'Humidity':[62],
          'WindSpeed':[11]
        })

          prediction = model.predict(new_data)

          print("Predicted Energy Consumption:", prediction[0])
```

Predicted Energy Consumption: 452.6410998552822

In []: